

Tasks:

- 1. Install Nmap on your system.**
- 2. Verify installation:**
 - `nmap --version`
- 3. Scan your localhost:**
 - `nmap 127.0.0.1`
- 4. Scan a practice target:**
 - `nmap scanme.nmap.org`
- 5. Perform the following scans:**
 - **Basic Scan**
 - **Service Version Scan (-sV)**
 - **OS Detection (-O)**
 - **Fast Scan (-F)**
- 6. Identify:**
 - **Open Ports**
 - **Running Services**
 - **Service Versions**
 - **Operating System (if detected)**
- 7. Take screenshots of all scan results.**

Assignment:

Create a report containing:

- 1. What is Nmap?**
- 2. Why is Nmap used?**
- 3. What is a Port?**
- 4. Difference between Open, Closed and Filtered Ports.**
- 5. Screenshot of each scan.**
- 6. List all detected services and ports.**
- 7. Your observations.**

Find the purpose of these ports:

- **21**
- **22**
- **25**
- **53**
- **80**
- **110**
- **143**
- **443**

- 3306
- 8080

Explain:

- Service Name
- Purpose
- Security Risks

Submission:

PDF or Word Document.

Aim

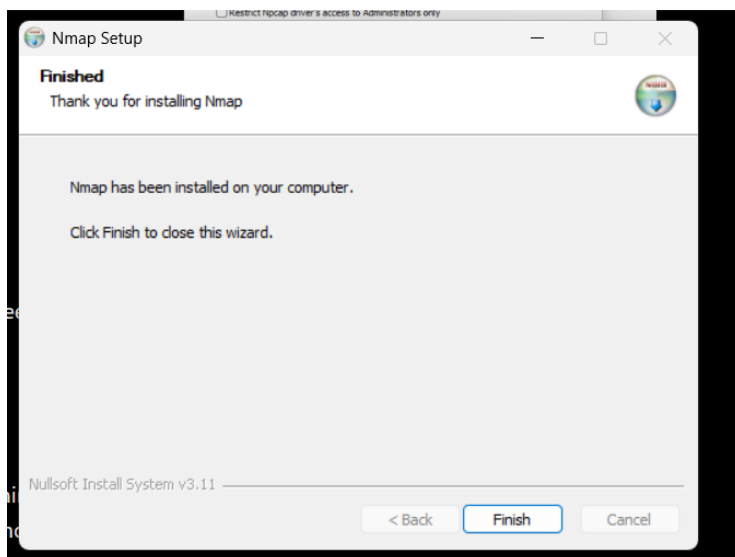
To install Nmap and perform different network scanning techniques to identify open ports, running services, service versions, and operating system information.

Step 1: Install Nmap

Windows

1. Visit [Nmap Official Website](https://nmap.org)
2. Download the latest Windows installer.
3. Run the installer.
4. Keep default settings.
5. Complete installation.

Screenshot 1



Step 2: Verify Installation

Open Command Prompt and type: nmap --version

Screenshot 2

```
C:\Users\vaish>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcre2-10.47 Npcap-1.87 nmap-
libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select
C:\Users\vaish>
```

Step 3: Basic Scan on Localhost

Command: `nmap 127.0.0.1`

Purpose:

- Scans local machine.
- Identifies open ports.

Screenshot 3

```
nmap 127.0.0.1

Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 12:47 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000078s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
902/tcp    open  iss-realservice
912/tcp    open  apex-mesh
3306/tcp   open  mysql

Nmap done: 1 IP address (1 host up) scanned in 0.73 seconds
```

Step 4: Scan Practice Target

Command: `nmap scanme.nmap.org`

Purpose:

- Scans Nmap's official testing server.
- Safe for learning.

Screenshot 4

```
nmap scanme.nmap.org

Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 12:49 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    open  http
135/tcp    filtered msrpc
139/tcp    filtered netbios-ssn
179/tcp    filtered bgp
445/tcp    filtered microsoft-ds
646/tcp    filtered ldp
9929/tcp   open  nping-echo
31337/tcp  open  Elite

Nmap done: 1 IP address (1 host up) scanned in 7.50 seconds
```

Step 5: Basic scan

Command: `nmap scanme.nmap.org`

Screenshot 5

```
nmap scanme.nmap.org

Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 12:49 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered   smtp
80/tcp    open       http
135/tcp    filtered   msrpc
139/tcp    filtered   netbios-ssn
179/tcp    filtered   bgp
445/tcp    filtered   microsoft-ds
646/tcp    filtered   ldp
9929/tcp   open       nping-echo
31337/tcp  open       Elite

Nmap done: 1 IP address (1 host up) scanned in 7.50 seconds
```

Step 6: Service Version Scan

Command: `nmap -sV scanme.nmap.org`

Purpose:

- Detects software running on ports.
- Displays version information.

Screenshot 6

```
nmap -sV scanme.nmap.org

Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 12:52 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open       ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered   smtp
80/tcp    open       http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp    filtered   msrpc
139/tcp    filtered   netbios-ssn
179/tcp    filtered   bgp
445/tcp    filtered   microsoft-ds
646/tcp    filtered   ldp
9929/tcp   open       nping-echo   Nping echo
31337/tcp  open       tcpwrapped

Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.10 seconds
```

Step 7: Operating System Detection

Command: `nmap -O scanme.nmap.org`

Purpose:

- Attempts OS fingerprinting.

Screenshot 7

```
nmap -O scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 12:54 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
179/tcp   filtered bgp
445/tcp   filtered microsoft-ds
646/tcp   filtered ldp
9929/tcp  open  sping-echo
31337/tcp open  Elite
Device type: general purpose|router|firewall
Running (JUST GUESSING): Linux 5.X|4.X|3.X|6.X|2.6.X (91%), MikroTik RouterOS 7.X (87%), SonicWALL embedded (85%)
OS_CPEs: cpe:/o:linux:linux_kernel:5.4 cpe:/o:linux:linux_kernel:4 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3 cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:6.1 cpe:/o:linux:linux_kernel:12.6.32 cpe:/o:sonicwall:aventail_ex-1500
Aggressive OS guesses: Linux 5.4 (91%), Linux 4.19 - 5.15 (88%), Linux 5.0 - 5.14 (87%), Linux 5.18 (87%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (87%), Linux 3.10 - 4.11 (86%), Linux 6.1 (86%), Linux 4.19 (86%), OpenWrt 21.02 (Linux 5.4) (85%), Linux 2.6.32 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 28 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 17.12 seconds
```

Step 8: Fast Scan

Command: `nmap -F scanme.nmap.org`

Purpose:

- Scans only common ports.
- Faster than full scan.

Screenshot 8

```
nmap -F scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 13:00 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.30s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 92 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
179/tcp   filtered bgp
445/tcp   filtered microsoft-ds
646/tcp   filtered ldp

Nmap done: 1 IP address (1 host up) scanned in 6.42 seconds
```

Result and records:

Scan Type	Open Ports	Running Services	Service Versions	Operating System
Localhost Scan	135/tcp 445/tcp 902/tcp 912/tcp 3306/tcp	msrpc microsoft-ds iss-realsecure apex-mesh mysql	No	Linux 5.4
Basic Scan	22/tcp 80/tcp 9929/tcp 31337/tcp	ssh http nping-echo Elite	No	Linux 5.4
Service Scan	22/tcp 80/tcp 9929/tcp 31337/tcp	ssh http nping-echo topwrapped	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0) Apache httpd 2.4.7 ((Ubuntu)) Nping echo	Linux 5.4
OS Scan	22/tcp 80/tcp 9929/tcp 31337/tcp	ssh http nping-echo Elite	NO	Linux 5.4
Fast Scan	22/tcp 80/tcp	ssh http	NO	Linux 5.4

Theory Questions

What is Nmap?

Nmap (Network Mapper) is an open-source network scanning and security auditing tool used to discover hosts, open ports, running services, operating systems, and vulnerabilities in a network.

Why is Nmap Used?

Nmap is used for:

- Network Discovery
- Security Auditing
- Port Scanning
- Service Enumeration
- Vulnerability Assessment
- Network Inventory
- Firewall Testing

What is a Port?

A port is a logical communication endpoint used by applications and services to send and receive network traffic.

Example:

- Web Server → Port 80
- HTTPS Website → Port 443
- SSH Server → Port 22

Difference Between Open, Closed and Filtered Ports

Port State	Meaning	Nmap Response	Example
Open	An application or service is actively listening on the port and accepting connections.	Nmap can communicate with the service successfully.	Port 80 (HTTP), Port 22 (SSH)
Closed	No application is listening on the port, but the host is reachable.	Nmap receives a response indicating the port is closed.	A server with no FTP service running on Port 21
Filtered	Nmap cannot determine whether the port is open or closed because a firewall, router, or security device is blocking the probe.	No response or blocked response is received.	Firewall blocking access to Port 22

Commonly Detected Ports

Port 21

Service: FTP (File Transfer Protocol)

Purpose:

Transfers files between systems.

Security Risks:

- Sends credentials in plain text.
- Vulnerable to brute-force attacks.

Port 22

Service: SSH (Secure Shell)

Purpose:

Secure remote administration.

Security Risks:

- Weak passwords.
- Brute-force attacks.

Port 25

Service: SMTP

Purpose:

Sending emails.

Security Risks:

- Spam relay abuse.
- Email spoofing.

Port 53

Service: DNS

Purpose:

Domain name resolution.

Security Risks:

- DNS amplification attacks.
- Cache poisoning.

Port 80

Service: HTTP

Purpose:

Website communication.

Security Risks:

- Data transmitted in plain text.
- Man-in-the-middle attacks.

Port 110

Service: POP3

Purpose:

Receiving emails.

Security Risks:

- Credentials may be exposed.
- Eavesdropping.

Port 143

Service: IMAP

Purpose:

Email synchronization.

Security Risks:

- Unauthorized mailbox access.
- Weak authentication.

Port 443

Service: HTTPS

Purpose:

Secure web communication.

Security Risks:

- SSL/TLS misconfiguration.
- Certificate issues.

Port 3306

Service: MySQL

Purpose:

Database communication.

Security Risks:

- Unauthorized database access.
- SQL attacks.

Port 8080

Service: Alternative HTTP Port

Purpose:

Web servers, proxies, testing applications.

Security Risks:

- Exposed admin panels.
- Misconfigured web applications.

Observations

1. Nmap successfully discovered active hosts and services.
2. Different scan types provide different levels of information.
3. Service Version Scan reveals software versions running on open ports.
4. Fast Scan reduces scan time by checking only common ports.
5. OS Detection attempts to identify the target operating system.
6. Open ports indicate available services on the target machine.
7. Filtered ports may indicate firewall protection.
8. Nmap is a valuable tool for network reconnaissance and security assessment.

Conclusion

The experiment demonstrated the installation and use of Nmap for network scanning. Different scan techniques such as Basic Scan, Service Version Scan, OS Detection, and Fast Scan were performed successfully. Open ports, services, and operating system information were identified, showing how Nmap can be used for network administration and cybersecurity assessments.